Aarav J. Samal

+1 (914) 356-0078 | aaravsamal@gmail.com | https://aaravsmall.github.io

Education

Purdue University West Lafayette, IN

B.S. in Computer & Information Technology; John Martinson Honors College *Expected May 2027*

- **Relevant Coursework:** Enterprise Data Management, IT Architectures, Cybersecurity Fundamentals

Experience

AI Engineer Intern June 2026 – Aug. 2026

Bristol Myers Squibb

- Embedded directly with a **cross-functional team** of drug-discovery scientists, product, and engineering to scope and ship **30+ features** on a full-stack **Claude Agent SDK** platform (**React, TypeScript, Node.js, Express**) – owning each from ambiguous problem framing through production deployment.
- Designed and deployed a **Claude-driven self-service support system** (custom SSE endpoint, intent detection, autofix pipeline) that let non-technical scientists resolve bugs/feature requests without engineering involvement, cutting manual triage load and turnaround time from pain point to fix.
- Built a **self-learning observability layer** and startup preflight checks that surfaced extraction health across fast/slow pipeline cycles, catching silent runtime failures before data loss and preventing downstream disruption to scientists' workflows.

Mobile & Web App Developer Intern Aug. 2025 – Oct. 2025

Niramai Thermalytix

- Built a cross-platform mobile app enabling **offline image quality verification**, reducing screening delays for end users operating in low-connectivity clinical environments.
- Integrated a thermal breast cancer screening **AI model** into the web application, developed supporting **Node.js** backend services, and deployed components on **AWS and Azure** for scalable, secure screening workflows.

Co-President May 2024 – July 2026

PrepTech Foundation

- Co-led a student-run education nonprofit, managing hiring and operations for **30+ student employees** across instruction, outreach, and curriculum delivery.
- Scaled the organization's first session to **100+ students across 10+ states** through targeted outreach and strategic partnerships.

Projects

Britney.ai | *Python, FastAPI, Postgres, Flutter, OpenAI API*

- Built and deployed an autonomous trading agent (**Python, FastAPI, Postgres**) running unattended 24/7 on a **DigitalOcean** droplet, executing risk-tiered portfolio rebalancing across 3 strategies with live discovery, classification, and emergency stop-loss systems.
- Integrated **OpenAI GPT-4o-mini** into a tool-calling chat assistant grounded in real trade/cash-ledger data, and shipped a cross-platform **Flutter** frontend backed by a **190-test pytest suite** running in GitHub Actions CI.
- Independently scoped system architecture with no fixed spec – deciding build order, risk tiers, and stop-loss logic from first principles – then validated each design choice against live production trades before scaling further.

Deki.ai | *TypeScript, Express, Electron, whisper.cpp, Ollama*

- Built a local-only AI desktop overlay (**Electron**) that captures the screen and system audio, transcribes speech with **whisper.cpp**, and streams answers from local **Ollama** models with no cloud APIs.
- Designed a from-scratch **TypeScript/Express** server with a two-stage vision-then-reasoning pipeline, automatic model fallback, and streamed responses, shipped with a typecheck/lint/test **GitHub Actions** pipeline.

Technical Skills

Languages: TypeScript, JavaScript, Python, Dart, SQL, Java, C++, C#
Frameworks & Libraries: React, Node.js, Express, Flutter, Vite, Tailwind CSS, Vitest, FastAPI, SQLAlchemy
AI & Agents: Claude Agent SDK, Model Context Protocol (MCP), OpenAI API (GPT-4o-mini), LLM tool use, prompt engineering, semantic similarity search
Cloud & Infrastructure: AWS, Azure, GCP, Docker, GitHub Actions, Git, DigitalOcean, systemd, Linux (Ubuntu)
Tools & APIs: Jira, MySQL, Postgres, SQLite, REST APIs, WebSockets, Server-Sent Events, Yahoo Finance API, Finnhub API
Testing: pytest, Hypothesis (property-based testing), Playwright (E2E)